



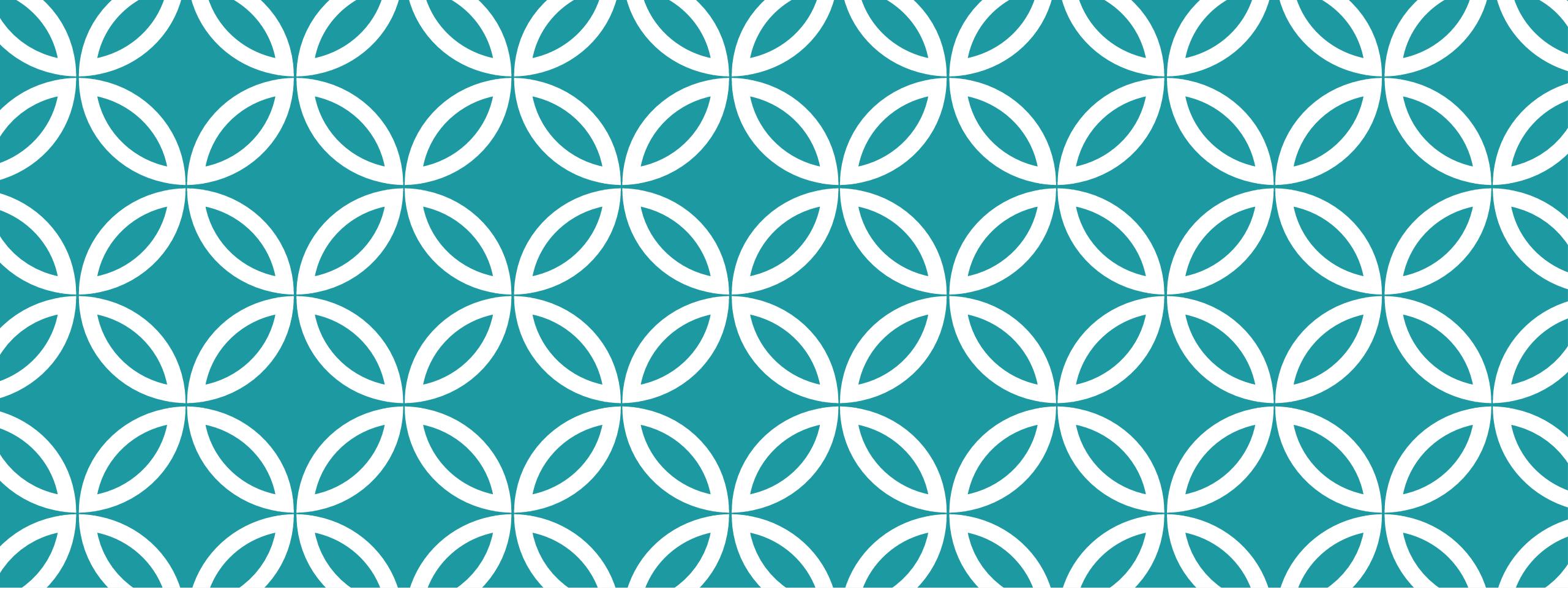
FLAME
UNIVERSITY

INTRODUCTION TO DEMOGRAPHY (POPULATION AND SOCIETY)

PUBP 233

Week 5: Morality measures

Shivakumar Jolad



AGE SPECIFIC MORTALITY |

BASIC MEASURES OF MORTALITY

The crude death rate (CDR)

$$\text{CDR}[0, T] = \frac{\text{Number of Deaths } [0, T]}{\text{PY}[0, T]}$$

Age-specific mortality rates

$${}_n m_x = \frac{{}_n D_x}{N_x} \cdot 1000$$

where:

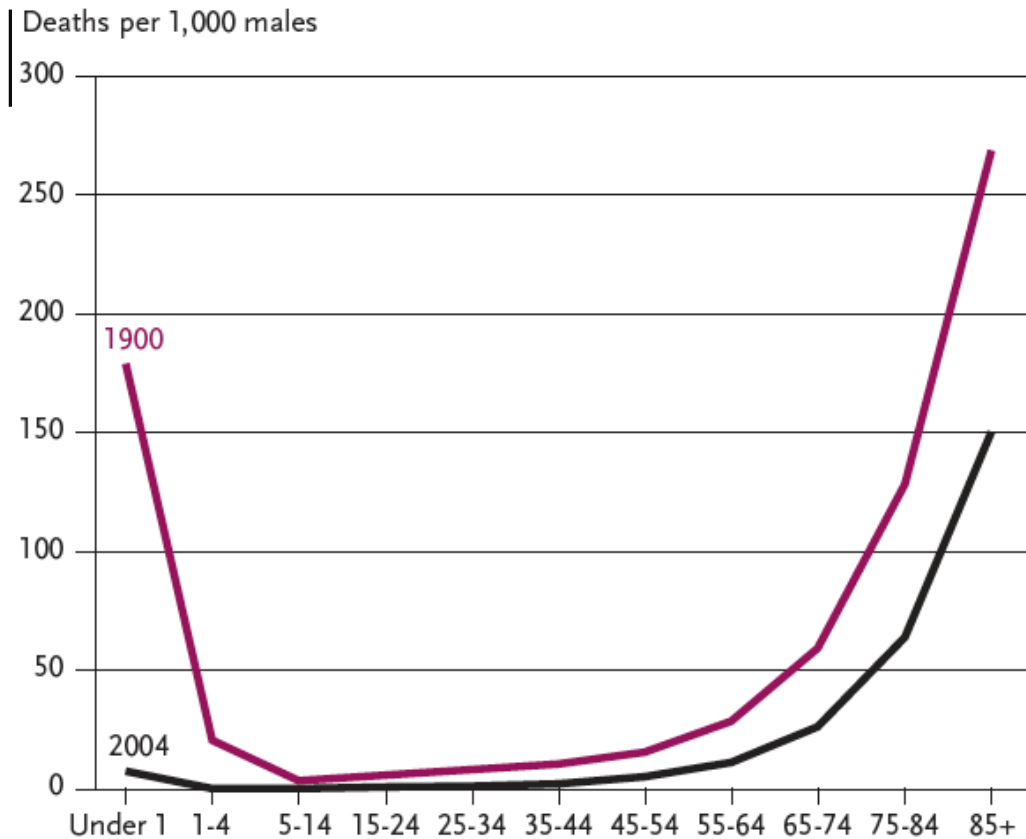
${}_n D_x$ = the number of deaths between ages x and $x+n$ in the year

${}_n N_x$ = the (mid-year) population aged between x and $x+n$.

For ease of interpretation, these rates are typically expressed per 1000, or sometimes per 100 000.

DEATH RATES

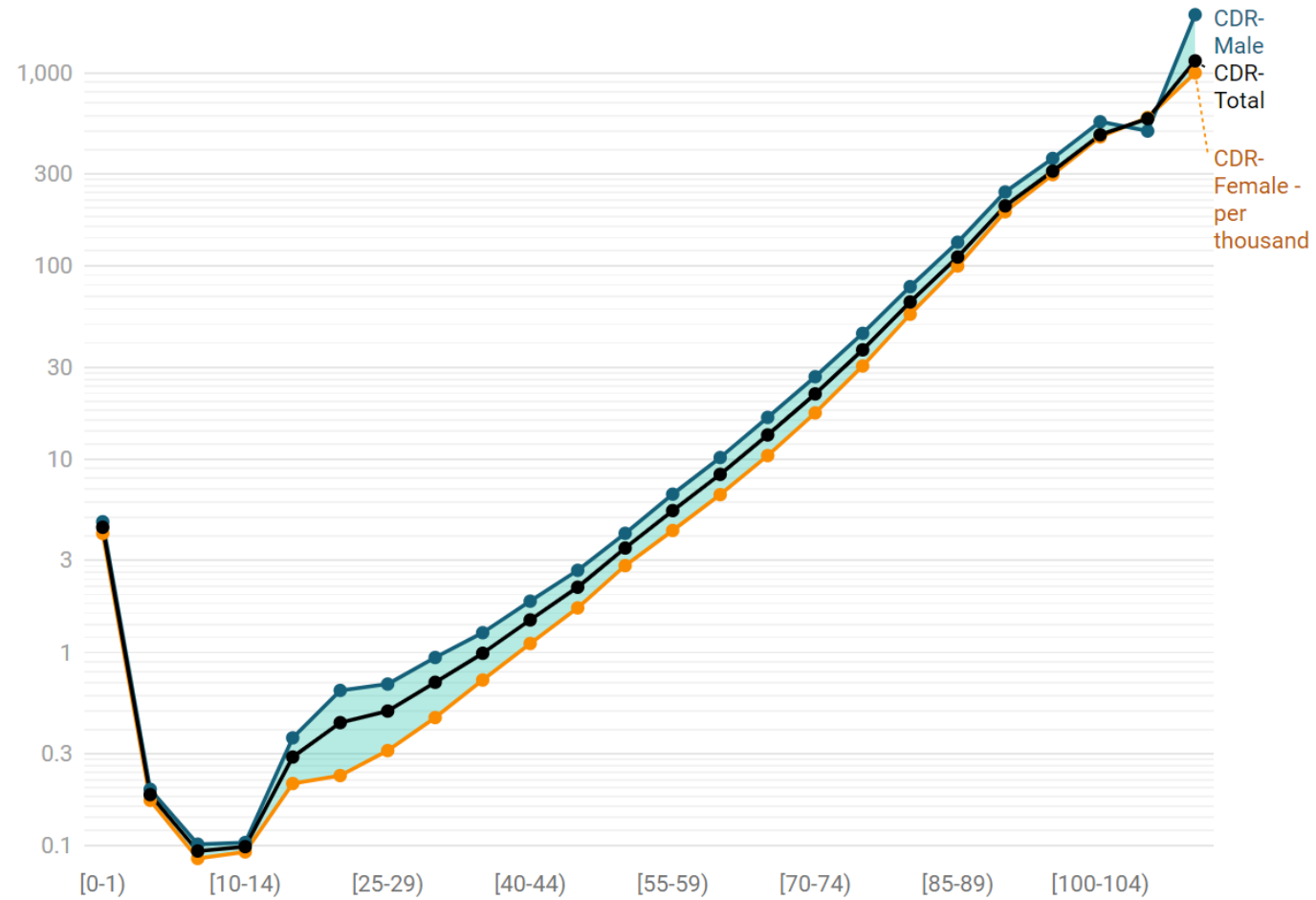
Death Rates for U.S. Males, by Age, 1900 and 2004



Source: U.S. Bureau of the Census, *Sixteenth Census of the United States: 1940, Vital Statistics Rates in the United States 1900-1940*, (1943): table 5; and National Center for Health Statistics, National Vital Statistics System, *Mortality*, 2004 (forthcoming).

McFalls, Joseph A. *Population: a lively introduction*. (PRB, 2007)

Age Specific Death Rates UK-2010



[Get the data](#) • Created with [Datawrapper](#)

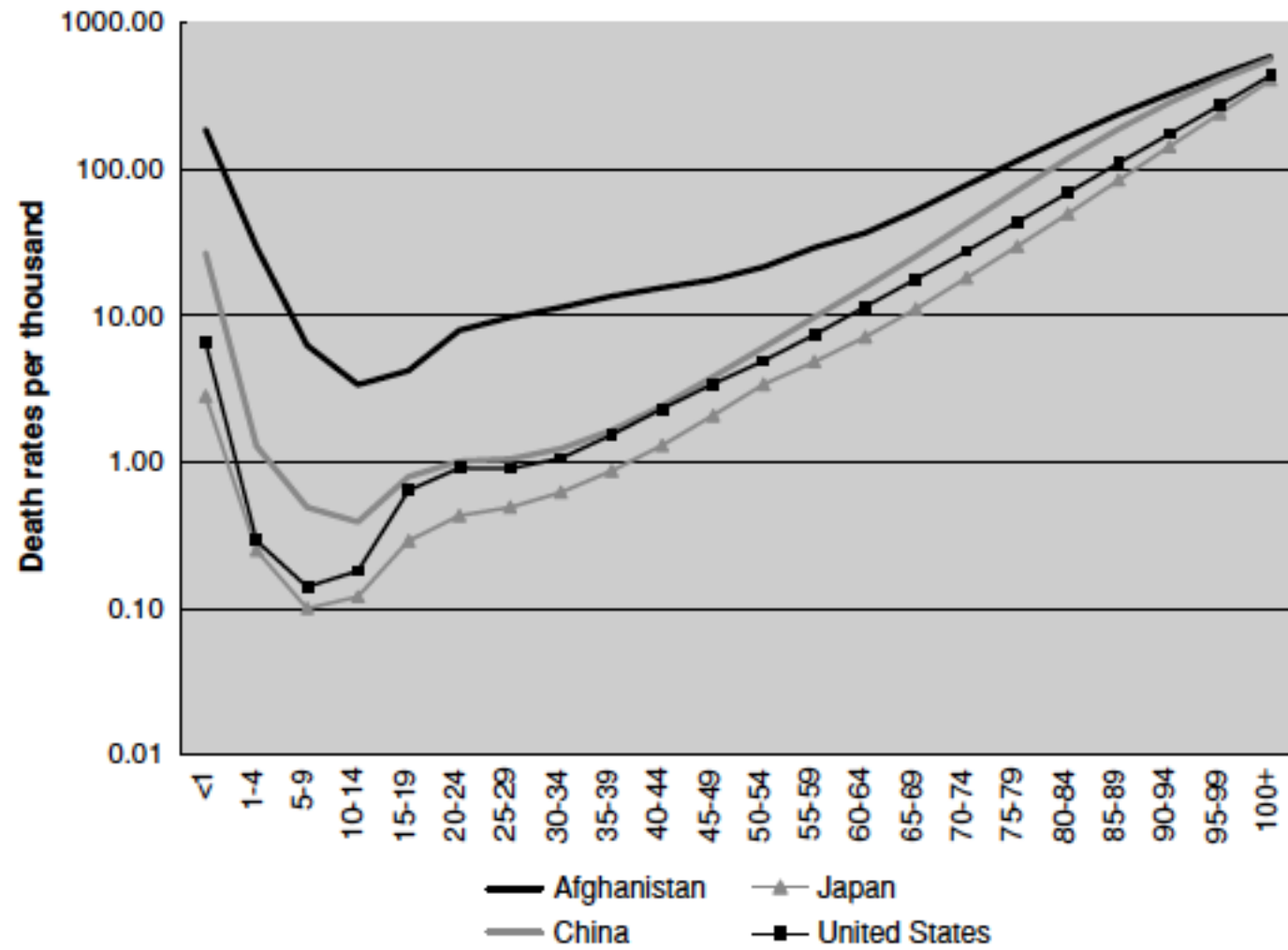
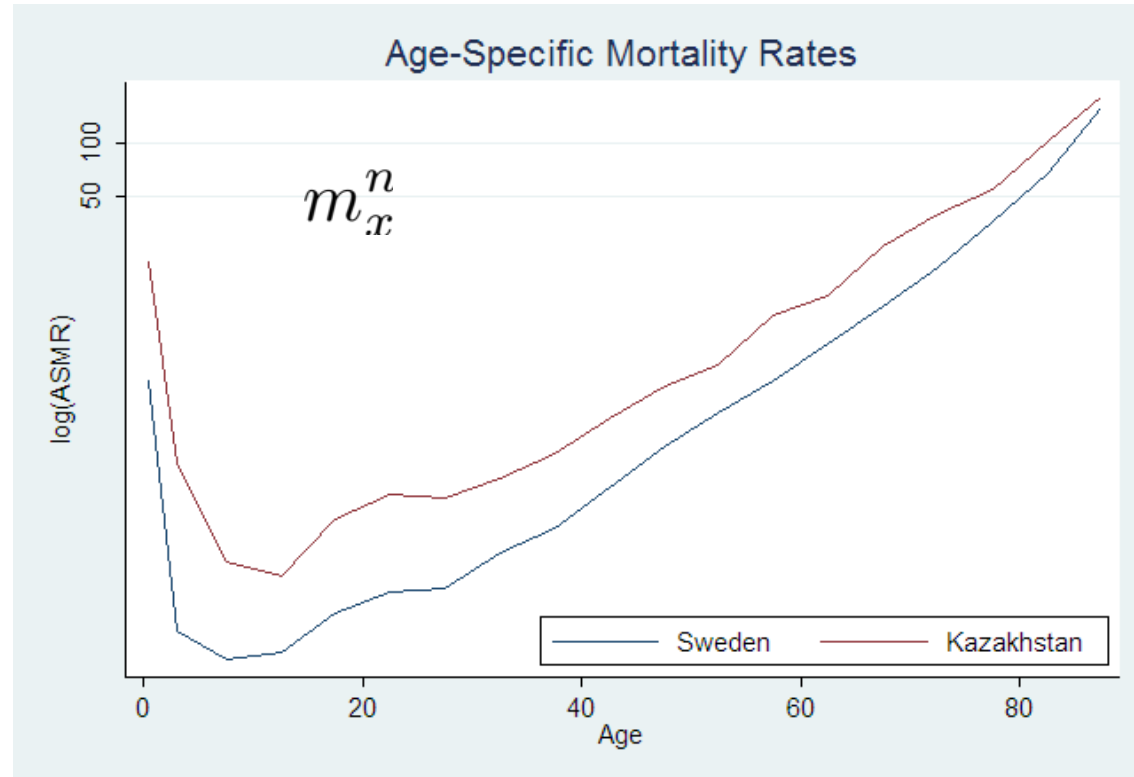


Figure 5.1. Age curves of mortality: Afghanistan, China, United States, and Japan, 2005.

AGE SPECIFIC VITAL RATES- MORTALITY



<http://data.princeton.edu/eco572/std.html>

m_x^n

Measures Deaths in a specific age group $[x, x+n)$ divided by the mid-point population of that age group

Why Women Live Longer than Men

by Thomas T. Perls, M.D., M.P.H.
Harvard Medical School

Ruth C. Fretts, M.D., M.P.H.
Harvard Medical School

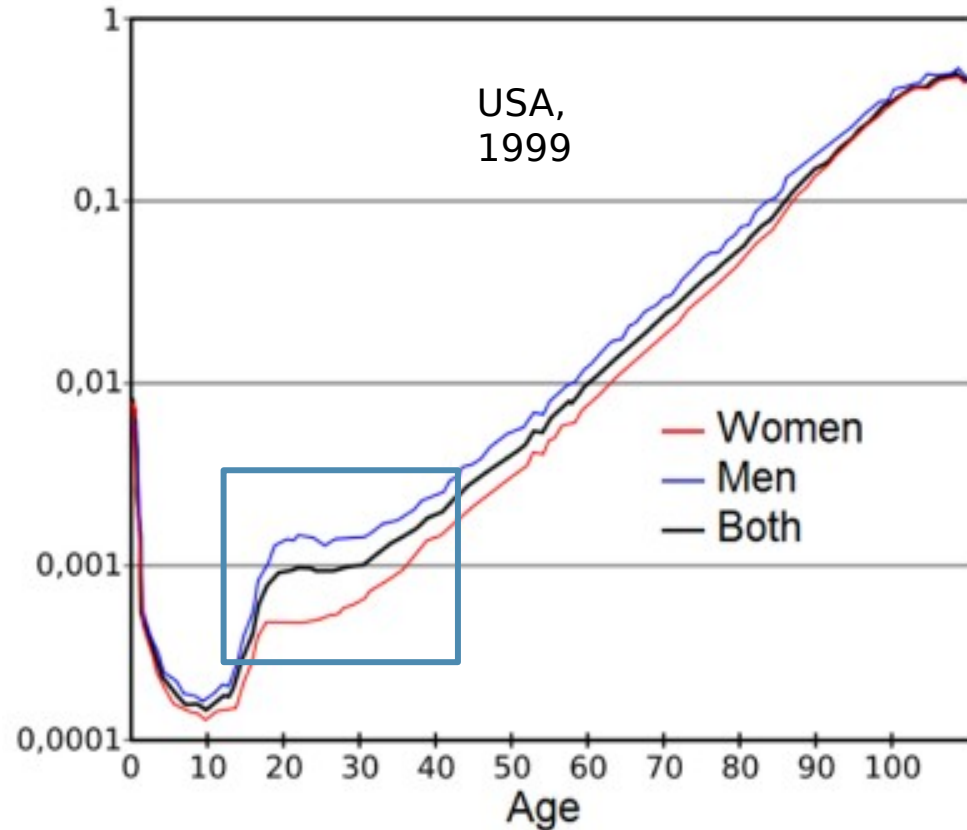
Women around the world have
a survival advantage over men—
sometimes by as much as 10 years.
What gives them the upper hand?

100 SCIENTIFIC AMERICAN PRESENTS

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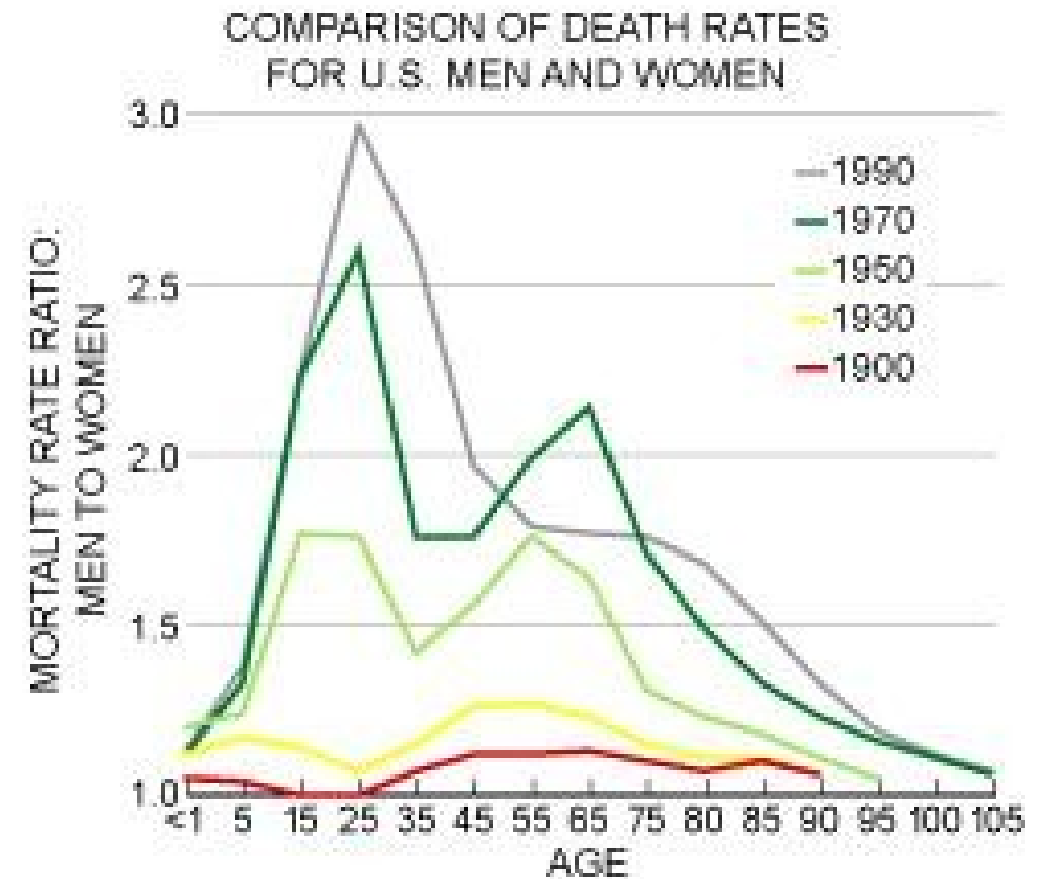


MALE- FEMALE DIFFERENCES IN MORTALITY

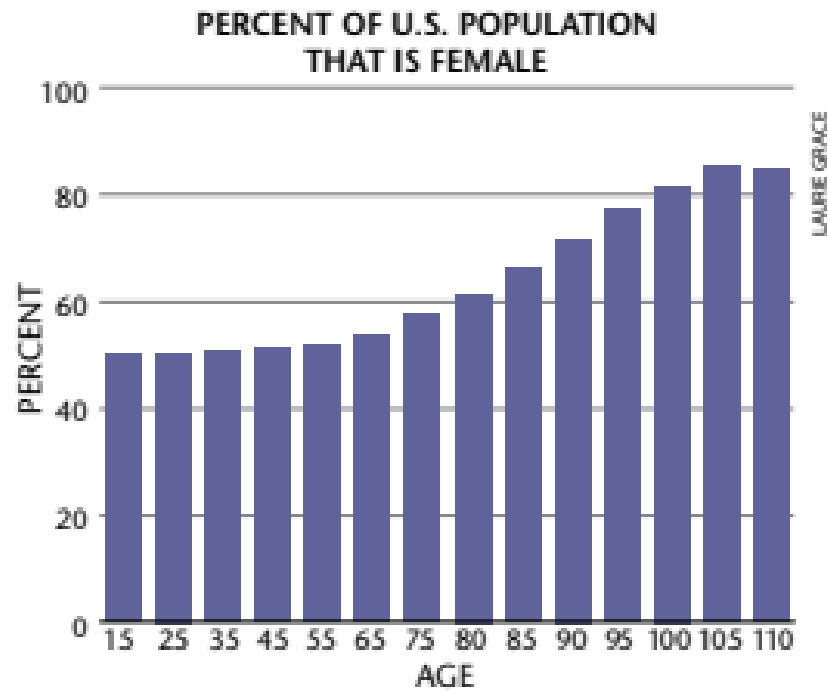


Age-related mortality in the United States in 1999.
Data from

http://www.programmed-aging.org/human_mortality_data.pdf

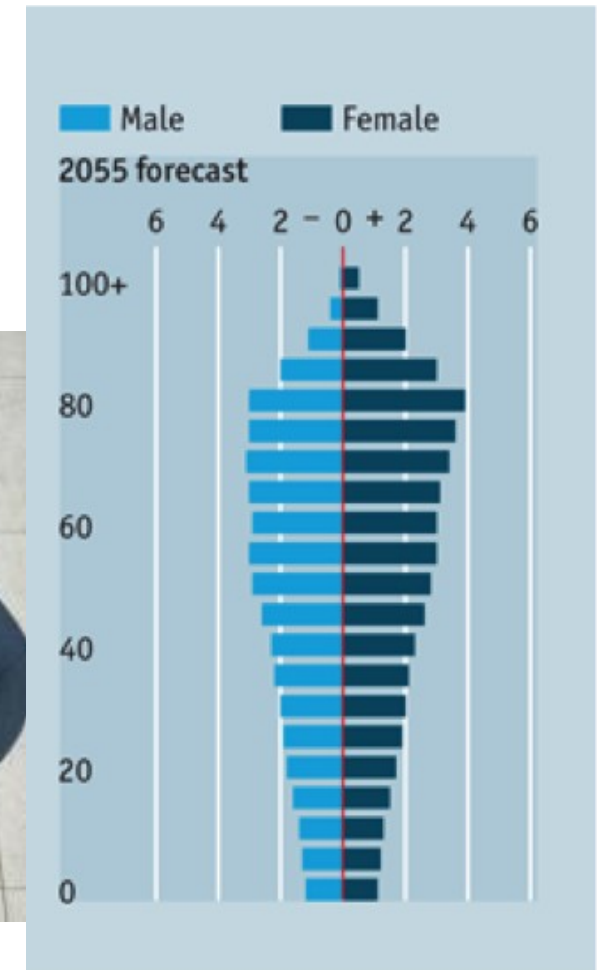


Why Women Live Longer than Men Thomas Perls & Ruth Fretts ,
https://www.cmu.edu/CSR/case_studies/women_live_longer.html



Women outnumber men by age 25, when they make up 50.3 percent of the U.S. population; by age 100, women comprise 81.7 percent.

SOURCE: Social Security Administration (1990 data)



JAPAN
2055

If female longevity is the product of evolutionary forces, then one might wonder what physiological mechanisms have evolved to support the preferential survival of women over men.

CORE PATTERN- MALE AND FEMALE MORTALITY RATES AND LIFE EXPECTANCY

- Across almost all societies, **women outlive men**—often by **5–10 years**.
- In the U.S., life expectancy at birth is ~79 years for women vs ~72 for men.
- The gap widens with age: among centenarians, women outnumber men **9:1**.
- Male mortality exceeds female mortality **at every age**, including infancy and childhood.
- **Early life:** Higher male fetal loss, infant mortality, and childhood deaths.
- **Adolescence & young adulthood (15–24):** Male death rates spike due to accidents, violence, suicide, and risk-taking behavior.
- **Midlife (40s–60s):** Men experience a sharp rise in heart disease and stroke; women's risk rises mainly after menopause.
- Behavioral causes dominate early; **chronic disease dominates later**.

ROLE OF HORMONES: TESTOSTERONE VS ESTROGEN

Testosterone is linked to:

- ▢ Risk-taking and aggression (behavioral risk)
- ▢ Higher LDL (“bad” cholesterol) and lower HDL (“good” cholesterol)

Estrogen:

- ▢ Improves cardiovascular profiles
- ▢ Acts as an antioxidant
- ▢ May delay onset of heart disease and Alzheimer’s

Evidence is suggestive, not definitive—but hormonally mediated risks are plausible.

OTHER FACTORS

Living Longer ≠ Being Healthier

- Women suffer more **chronic but nonfatal conditions** (arthritis, osteoporosis, autoimmune diseases).
- Men are more likely to die from **acute and fatal conditions** (heart disease, cancer).
- This suggests women often *live with disease*, while men *die from disease*
- Female longevity is observed across species (e.g., primates, whales).

Evolutionary Explanation:

- Evolution favors longer-lived females because:
- Human children require long-term care.
- Women's reproductive success depends on survival over many decades.
- Male reproductive success is less dependent on longevity than access to mates.

BIOLOGICAL FACTORS

Menopause as an Evolutionary Advantage

Menopause reduces maternal mortality from late-age childbirth.

Post-reproductive women enhance offspring survival via caregiving (“grandmother effect”).

Longevity may be a **by-product of genes promoting slow aging and extended fertility**.

Biological Mechanisms Supporting Longevity

Lower lifetime iron levels in women → fewer oxidative reactions. Slower metabolic rates associated with longer lifespan.

Two X chromosomes offer genetic redundancy and better DNA repair capacity.

Some longevity-related genes are located on the X chromosome.

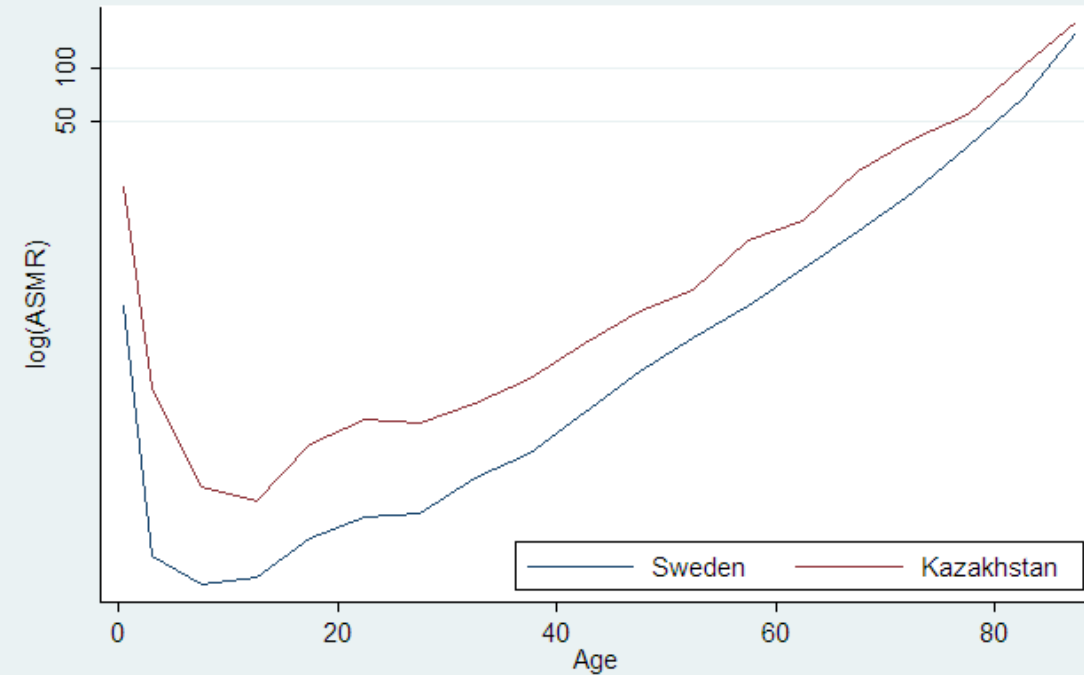
CRUDE DEATH RATE AND ASDR

Table 2.1: Comparison of crude death rates and age-specific death rates in two populations

Sweden, females, 1992					Kazakhstan, females, 1992				
Age group i	Mid-year population	Deaths during year	Death rate	Proportion in age category	Age group i	Mid-year population	Deaths during year	Death rate	Proportion in age category
N_i^{Sw}	D_i^{Sw}	M_i^{Sw}	C_i^{Sw}		N_i^K	D_i^K	M_i^K	C_i^K	
0	59,727	279	0.00467	0.0136	0	174,078	3,720	0.02137	0.0200
1-4	229,775	42	0.00018	0.0524	1-4	754,758	1,220	0.00162	0.0868
5-9	245,172	31	0.00013	0.0559	5-9	879,129	396	0.00045	0.1011
10-14	240,110	33	0.00014	0.0548	10-14	808,510	298	0.00037	0.0929
15-19	264,957	61	0.00023	0.0604	15-19	720,161	561	0.00078	0.0828
20-4	287,176	87	0.00030	0.0655	20-4	622,988	673	0.00108	0.0716
25-9	311,111	98	0.00032	0.0709	25-9	733,057	752	0.00103	0.0843
30-4	280,991	140	0.00050	0.0641	30-4	732,312	965	0.00132	0.0842
35-9	286,899	197	0.00069	0.0654	35-9	612,825	1,113	0.00182	0.0704
40-4	308,238	362	0.00117	0.0703	40-4	487,996	1,405	0.00288	0.0561
45-9	320,172	643	0.00201	0.0730	45-9	284,799	1,226	0.00430	0.0327
50-4	242,230	738	0.00305	0.0552	50-4	503,608	2,878	0.00571	0.0579
55-9	210,785	972	0.00461	0.0481	55-9	301,879	3,266	0.01082	0.0347
60-4	216,058	1,640	0.00759	0.0493	60-4	374,317	5,212	0.01392	0.0430
65-9	224,479	2,752	0.01226	0.0512	65-9	256,247	6,866	0.02679	0.0295
70-4	222,578	4,509	0.02026	0.0508	70-4	154,623	6,182	0.03998	0.0178
75-9	184,102	6,745	0.03664	0.0420	75-9	149,917	8,199	0.05469	0.0172
80-4	140,667	9,587	0.06815	0.0321	80-4	88,716	9,013	0.10159	0.0102
85+	110,242	17,340	0.15729	0.0251	85+	58,940	10,627	0.18030	0.0068
All	4,385,469	46,256	0.01055	1.0000	All	8,698,860	64,572	0.00742	1.0000
CDR		10.55 p. 1,000			CDR		7.42 p. 1,000		

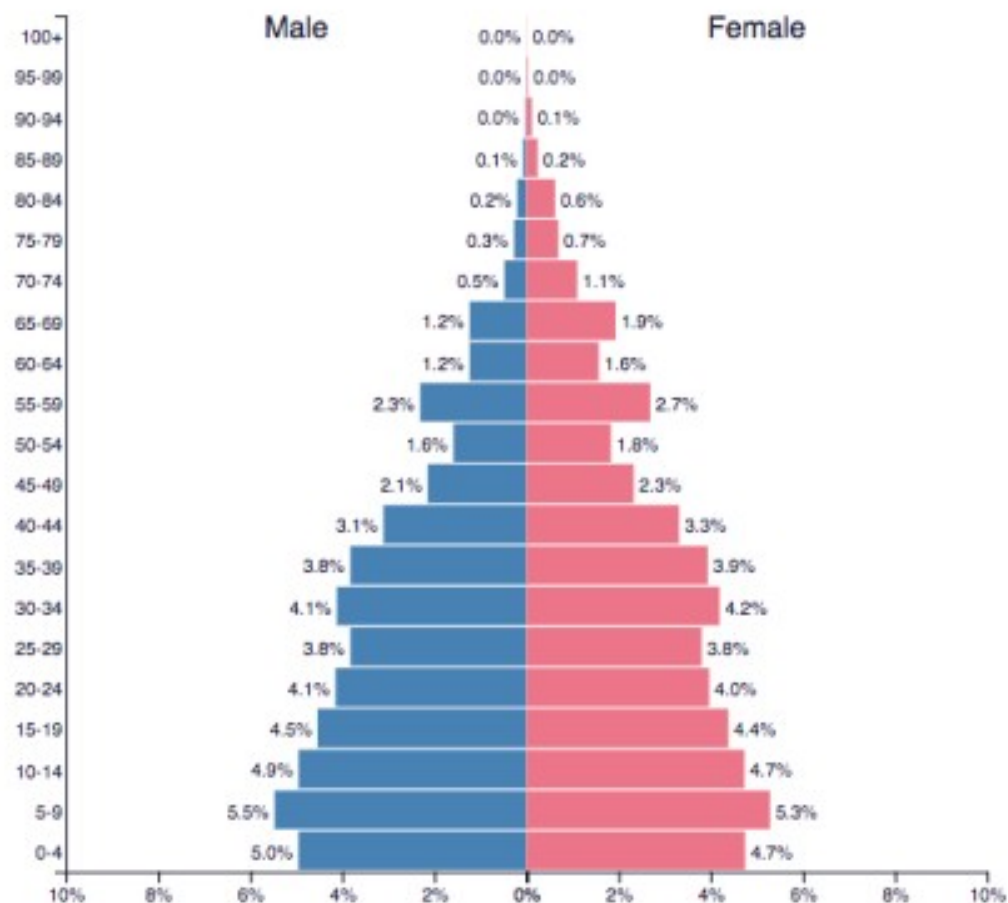
Data source: United Nations, Demographic Yearbook (various years).

Age-Specific Mortality Rates



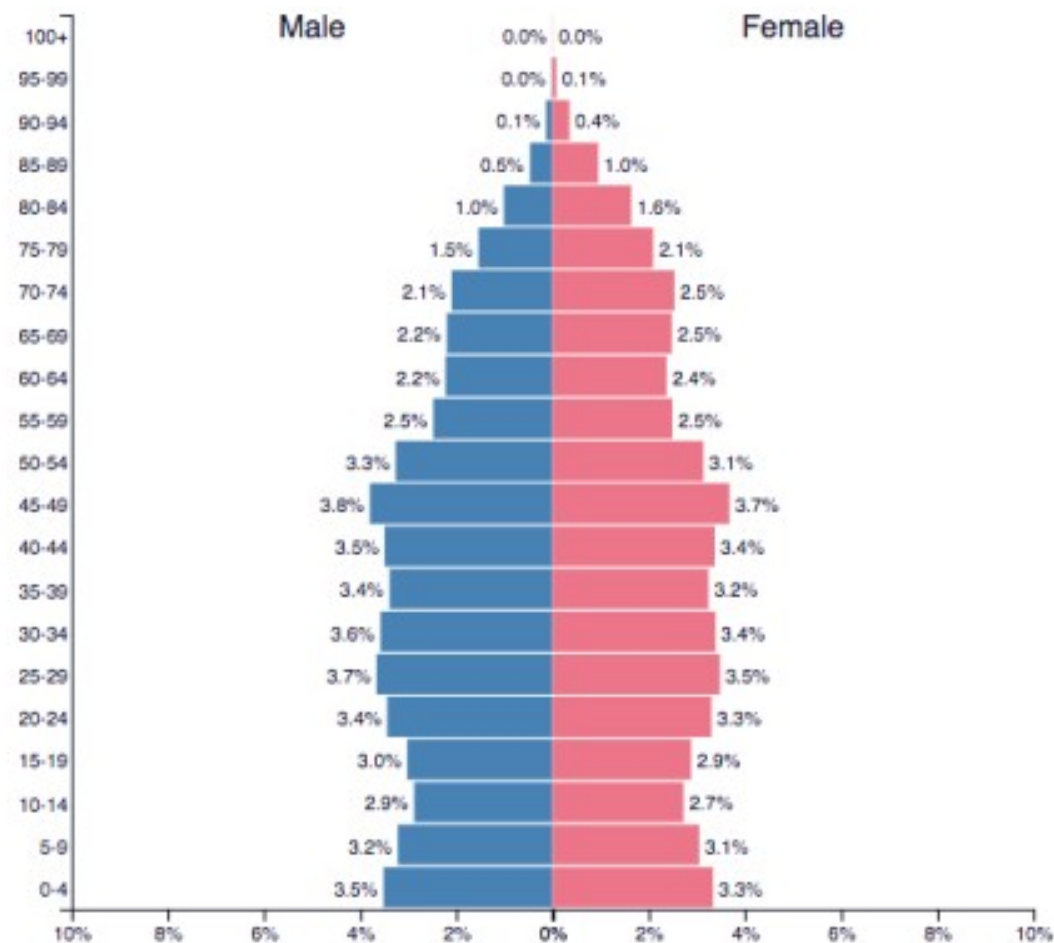
Kazakhstan ▼ 1994

Population: 16,022,224



Sweden ▼ 1994

Population: 8,798,234



EXERCISE ON POPULATION PYRAMID+ DEATH RATES+ AGE STANDARDIZATION



Sweden and Ecuador. In 1994 the crude death rate in Sweden was 10.5 per 1000 person-years, and in Ecuador was 5.6 per 1000 person-years.

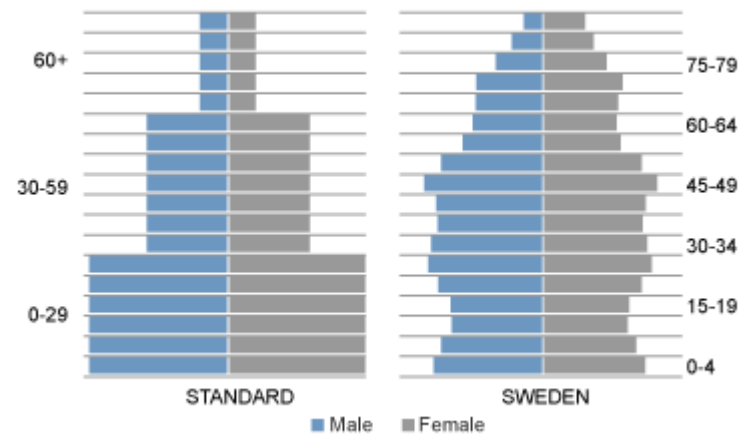
ECUADOR			
Age (Years)	No. of deaths	Population (1,000's)	Rate (per 1000 pyrs)
0-29	16,700	7,498	2.2
30-59	11,900	3,219	
60+	36,100	742	
All ages	64,700	11,459	5.6

SWEDEN			
Age (Years)	No. of deaths	Population (1,000's)	Rate (per 1000 pyrs)
0-29		3,397	0.45
30-59		3,475	2.4
60+	82,500	1,945	42.4
All ages	92,359	8,817	10.5

https://papp.iussp.org/sessions/papp101_s06/PAPP101_s06_050_030.html

The age distribution of an (imaginary) standard population

Age (years)	Population
0-29	56,000
30-59	33,000
60+	11,000
Total	100,000



Age (Years)	Mortality Rate for Sweden (per 1000 pyrs)	Standard Population	Expected No. of Deaths
0-29	0.45	56,000	25.2
30-59	2.4	33,000	
60+	42.4	11,000	

Age (years)	Mortality Rate for Sweden (per 1000 pyrs)	Standard Population	Expected No. of deaths
0-29	0.45	56,000	25.2
30-59	2.4	33,000	79.2
60+	42.4	11,000	466.4
Total		100,000	570.8

Age adjusted mortality rate:

= 570.8 / 100,000

= 5.7 per 1000 pyrs

ECUADOR

Age (years)	Mortality Rate for Ecuador (per 1000 pyrs)	Standard Population	Expected No. of deaths
0-29	2.2	56,000	123.2
30-59	3.7	33,000	
60+	48.7	11,000	535.7
		100000	

Age (years)	Mortality Rate for Ecuador (per 1000 pyrs)	Standard Population	Expected No. of deaths
0-29	2.2	56,000	123.2
30-59	3.7	33,000	122.1
60+	48.7	11,000	535.7
		100000	781

Sweden

- Age-adjusted mortality rate = **5.7** per 1000 pyrs

Ecuador

- Age-adjusted mortality rate = **7.8** per 1000 pyrs

- CMF = **7.8 / 5.7**
= **1.37**

Table 5.1. Age-specific death rates (ASDR) and age-specific proportions of the population: United States and Venezuela, 2005

Age group	United States		Venezuela	
	ASDR	Proportion of population	ASDR	Proportion of population
<1	0.00654	0.0139	0.01623	0.0219
1-4	0.00029	0.0548	0.00064	0.0800
5-9	0.00014	0.0675	0.00032	0.1038
10-14	0.00018	0.0719	0.00041	0.1023
15-19	0.00064	0.0724	0.00146	0.1012
20-24	0.00091	0.0702	0.00238	0.0925
25-29	0.00090	0.0671	0.00221	0.0842
30-34	0.00106	0.0679	0.00203	0.0726
35-39	0.00153	0.0706	0.00215	0.0688
40-44	0.00231	0.0766	0.00289	0.0630
45-49	0.00341	0.0757	0.00411	0.0515
50-54	0.00493	0.0675	0.00568	0.0437
55-59	0.00742	0.0575	0.00775	0.0346
60-64	0.01150	0.0437	0.01173	0.0251
65-69	0.01780	0.0339	0.01877	0.0180
70-74	0.02771	0.0284	0.02838	0.0136
75-79	0.04350	0.0248	0.04270	0.0097
80-84	0.06958	0.0187	0.07531	0.0053
85-89	0.11056	0.0107	0.12769	0.0024
90-94	0.17477	0.0045	0.20820	0.0007
95-99	0.27656	0.0013	0.32576	0.0001
100+	0.43892	0.0003	0.48975	0.0001

The United States in 2006 had a CDR of 8, while Venezuela had a CDR of 4. This means that there were twice as many deaths per 1,000 population in the United States in 2006 than there were in Venezuela.

CDR, ASDR AND AGE STRUCTURE

$$CDR = \sum nM_x \left(\frac{nP_x}{P} \right) * 1,000$$

P = total population,

nP_x = population in age group x, and

nM_x = ASDR for age group x (Palmore and Gardner, 1994: 15)

**CDR is the weighted average of
Age Specific Death Rate
M(x,n) with
Population age structure.
 $P(x,n)/P$**

Table 5.2. Age data for a hypothetical population

Age	Midyear population	Deaths	ASDR
0-34	2,000	40	20/1,000
35+	1,000	80	80/1,000
Total	3,000	120	40/1,000

Source: Palmore and Gardner, 1994: 15-16.

$$CDR = \frac{D}{P} * 1,000 = \frac{120}{3,000} * 1,000 = 0.04 * 1,000 = 40$$

$$CDR = \left[\frac{2,000}{3,000} * 20 \right] + \left[\frac{1,000}{3,000} * 80 \right] = \left[\frac{2}{3} * 20 \right] + \left[\frac{1}{3} * 80 \right]$$

$$CDR = \frac{40}{3} + \frac{80}{3} = \frac{120}{3} = 40$$

AGE STANDARDIZATION

$$ASCDR^{Sw} = \sum_{i=1}^{\infty} M_i^{Sw} \cdot C_i^s = \text{Age-standardized crude death rate for Sweden}$$

$$ASCDR^K = \sum_{i=1}^{\infty} M_i^K \cdot C_i^s = \text{Age-standardized crude death rate for Kazakhstan}$$

$$C_i^s = \left(\frac{C_i^{Sw} + C_i^K}{2} \right) = \text{Average age distribution}$$

KAZAKHSTAN AND SWEDEN

Age group i	Age distribution of Sweden C_i^{Sw}	Age distribution of Kazakhstan C_i^K	Average age distribution $\frac{C_i^{Sw} + C_i^K}{2}$	Age-specific death rate in Sweden M_i^{Sw}	Age-specific death rate in Sweden $M_i^{Sw} \cdot \frac{C_i^{Sw} + C_i^K}{2}$	Age-specific death rate in Kazakhstan M_i^K	Age-specific death rate in Kazakhstan $M_i^K \cdot \frac{C_i^{Sw} + C_i^K}{2}$
0	0.0136	0.0200	0.0168	0.00467	0.00008	0.02137	0.00036
1-4	0.0524	0.0868	0.0596	0.00018	0.00001	0.00162	0.00011
5-9	0.0559	0.1011	0.0785	0.00013	0.00001	0.00045	0.00004
10-14	0.0548	0.0929	0.0738	0.00014	0.00001	0.00037	0.00003
15-19	0.0604	0.0828	0.0716	0.00023	0.00002	0.00078	0.00006
20-4	0.0655	0.0716	0.0686	0.00030	0.00002	0.00108	0.00007
25-9	0.0709	0.0843	0.0776	0.00032	0.00002	0.00103	0.00008
30-4	0.0641	0.0842	0.0741	0.00050	0.00004	0.00132	0.00010
35-9	0.0654	0.0704	0.0679	0.00069	0.00005	0.00182	0.00012
40-4	0.0703	0.0561	0.0632	0.00117	0.00007	0.00288	0.00018
45-9	0.0730	0.0327	0.0529	0.00201	0.00011	0.00430	0.00023
50-4	0.0552	0.0579	0.0566	0.00305	0.00017	0.00571	0.00032
55-9	0.0481	0.0347	0.0414	0.00461	0.00019	0.01082	0.00045
60-4	0.0493	0.0430	0.0461	0.00759	0.00035	0.01392	0.00064
65-9	0.0512	0.0295	0.0403	0.01226	0.00049	0.02679	0.00108
70-4	0.0508	0.0178	0.0343	0.02026	0.00069	0.03998	0.00137
75-9	0.0420	0.0172	0.0296	0.03664	0.00108	0.05469	0.00162
80-4	0.0321	0.0102	0.0211	0.06815	0.00144	0.10159	0.00215
85+	0.0251	0.0068	0.0160	0.15729	0.00251	0.18030	0.00288
Sum	1.0000	1.0000	1.0000		0.00737		0.01188

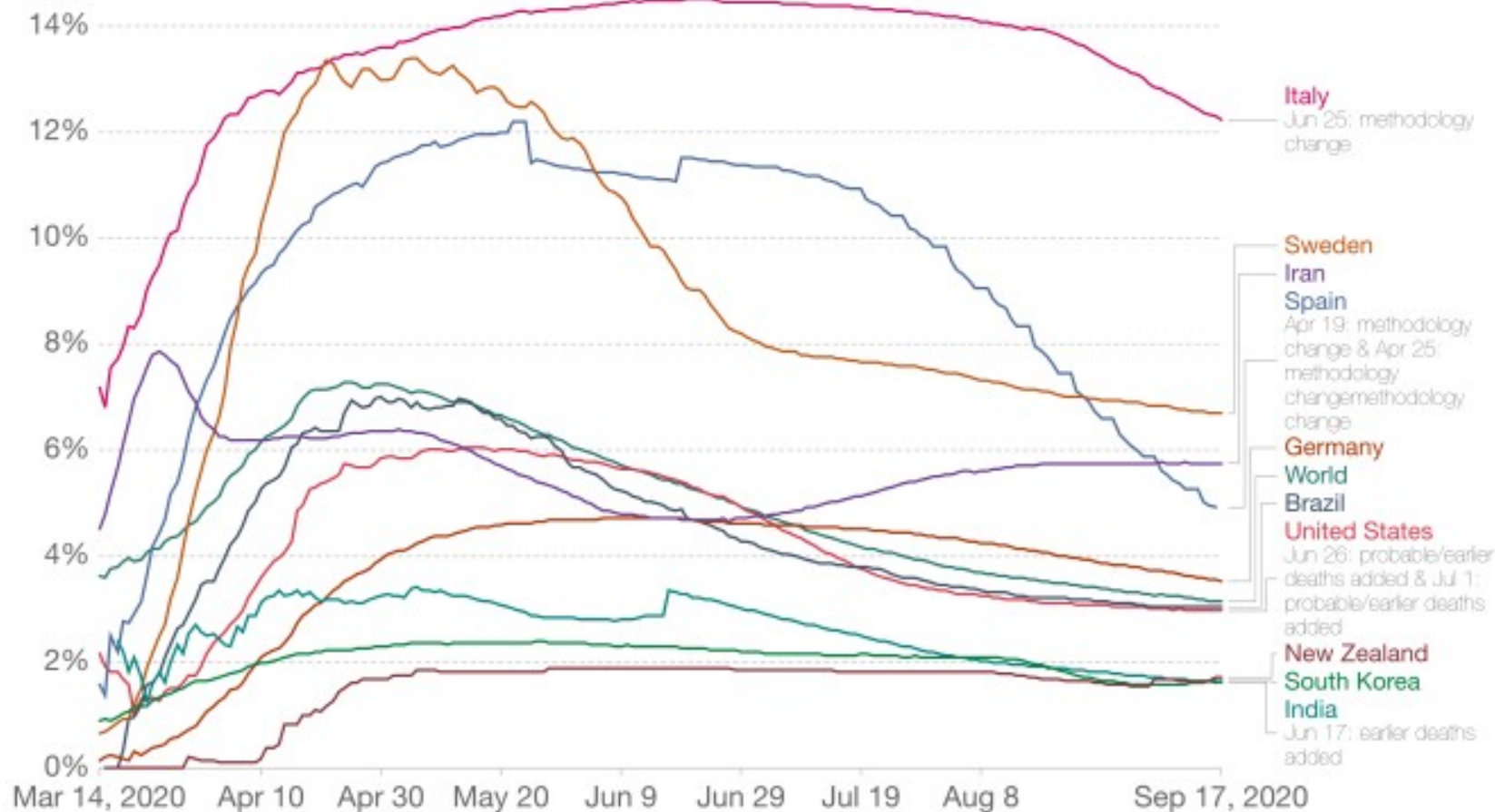
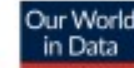
$ASCDR^{Sw} = 7.37 \text{ p. 1,000}$ $ASCDR^K = 11.88 \text{ p. 1,000}$

Data source: United Nations, *Demographic Yearbook* (various years).

COVID FATALITY

Case fatality rate of the ongoing COVID-19 pandemic

The Case Fatality Rate (CFR) is the ratio between confirmed deaths and confirmed cases. During an outbreak of a pandemic the CFR is a poor measure of the mortality risk of the disease. We explain this in detail at OurWorldInData.org/Coronavirus



Source: European CDC – Situation Update Worldwide – Last updated 16 September, 17:05 (London time)

CC BY

COVID FATALITY BY AGE

Coronavirus: case fatality rates by age

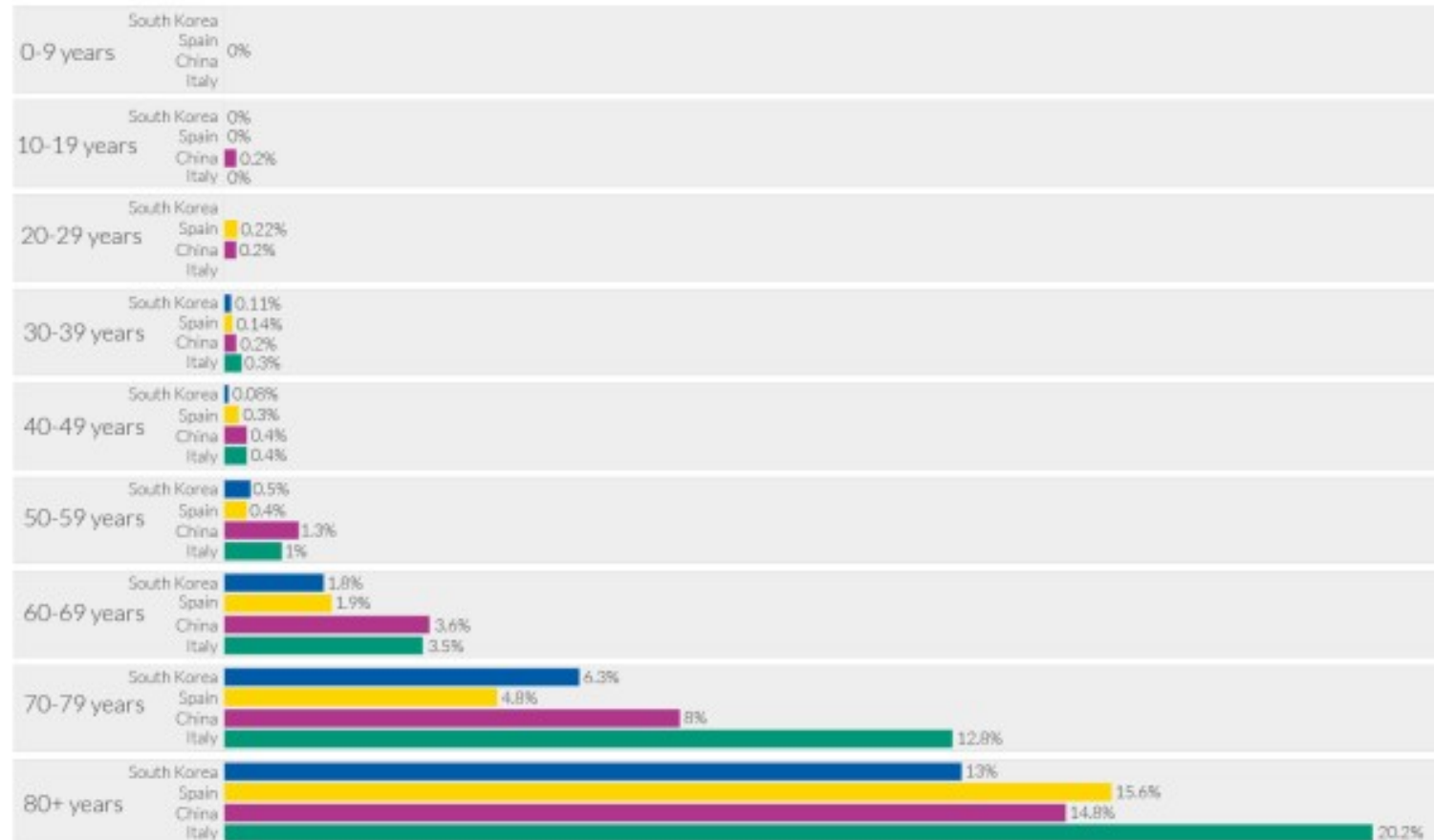
Our World
in Data

Case fatality rate (CFR) is calculated by dividing the total number of confirmed deaths due to COVID-19 by the number of confirmed cases.

Two of the main limitations to keep in mind when interpreting the CFR:

(1) many cases within the population are unconfirmed due to a lack of testing.

(2) some individuals who are infected will eventually die from the disease, but are still alive at time of recording.



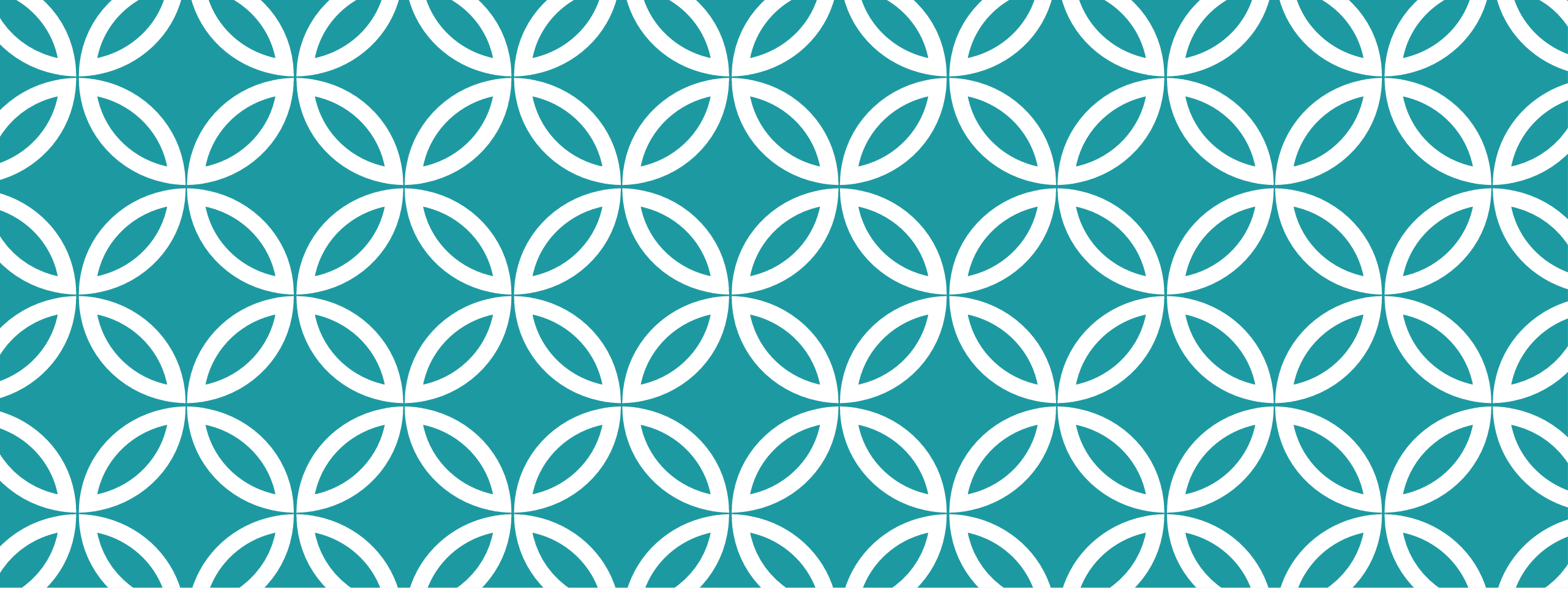
Note: Case fatality rates are based on confirmed cases and deaths from COVID-19 as of: 17th February (China); 24th March (Spain); 24th March (South Korea); 17th March (Italy).

Data sources: Chinese Center for Disease Control and Prevention (CDC); Spanish Ministry of Health; Korea Centers for Disease Control and Prevention (KCDC).

Order G, Rezza G, Brusaferro S. Case-Fatality Rate and Characteristics of Patients Dying in Relation to COVID-19 in Italy. JAMA.

OurWorldinData.org - Research and data to make progress against the world's largest problems.

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LEXIS DIAGRAM

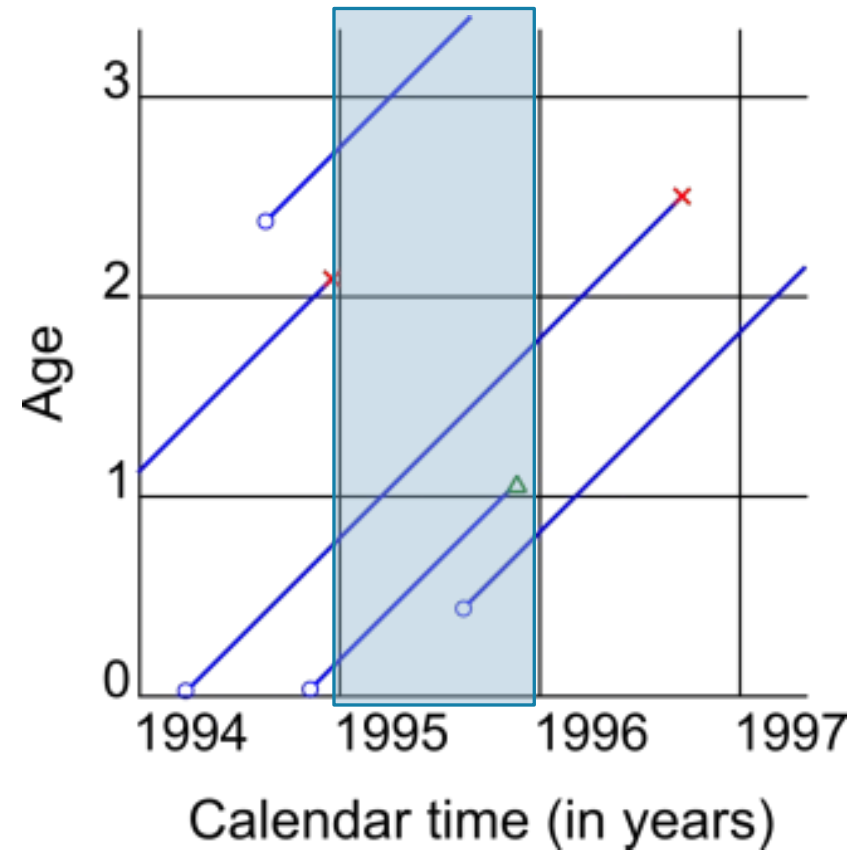
LEXIS DIAGRAM- PERIOD

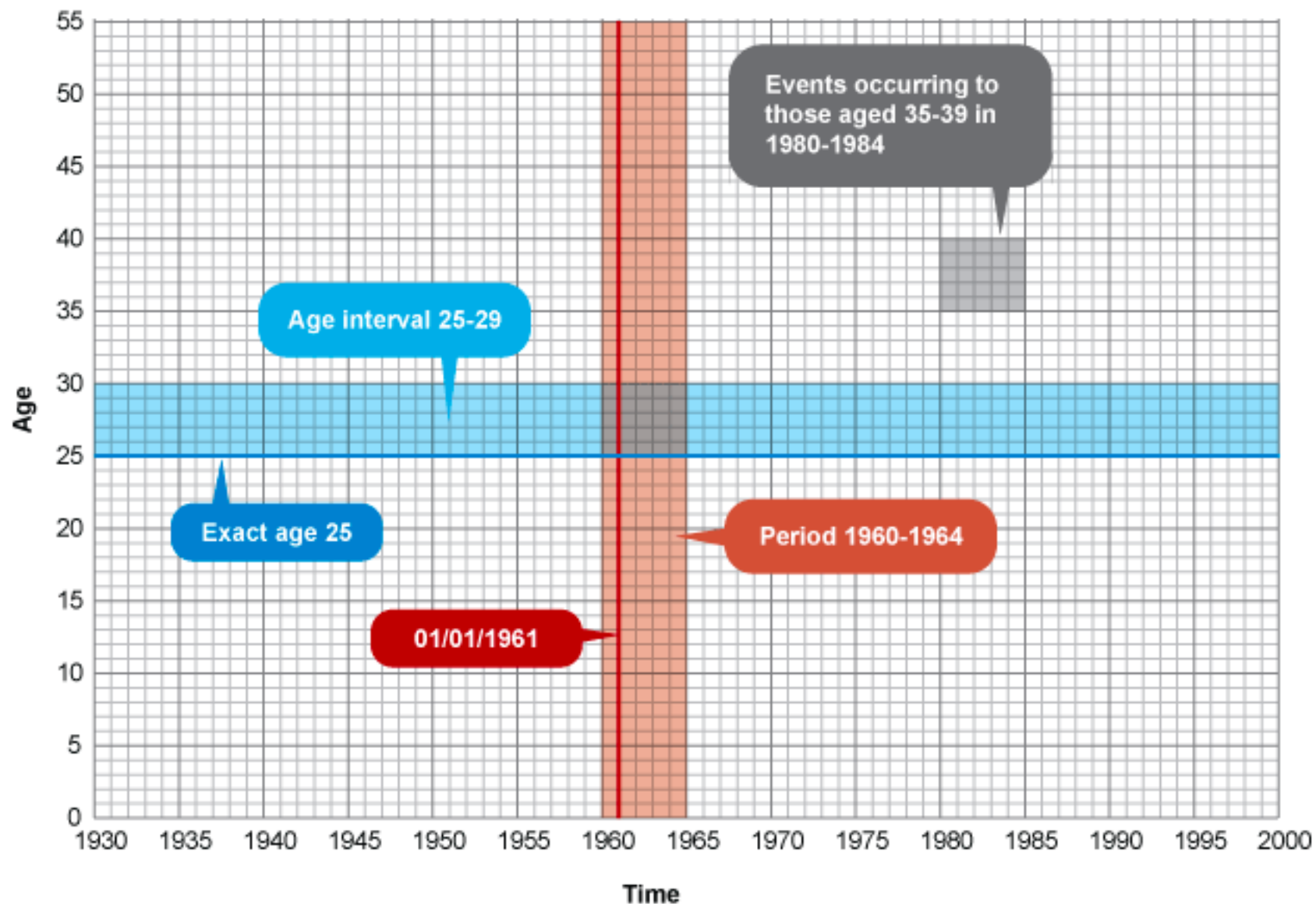
Vital events during a particular time period

Eg: Births and deaths between 1995-96

Literacy rate

Age group





Measures of age by calendar time (age-period measures) are represented by the square boxes within the grid. These show demographic events which occur to people within a specific age interval during a specific time interval.

http://papp.iussp.org/sessions/papp101_s02/PAPP101_s02_100_020.html

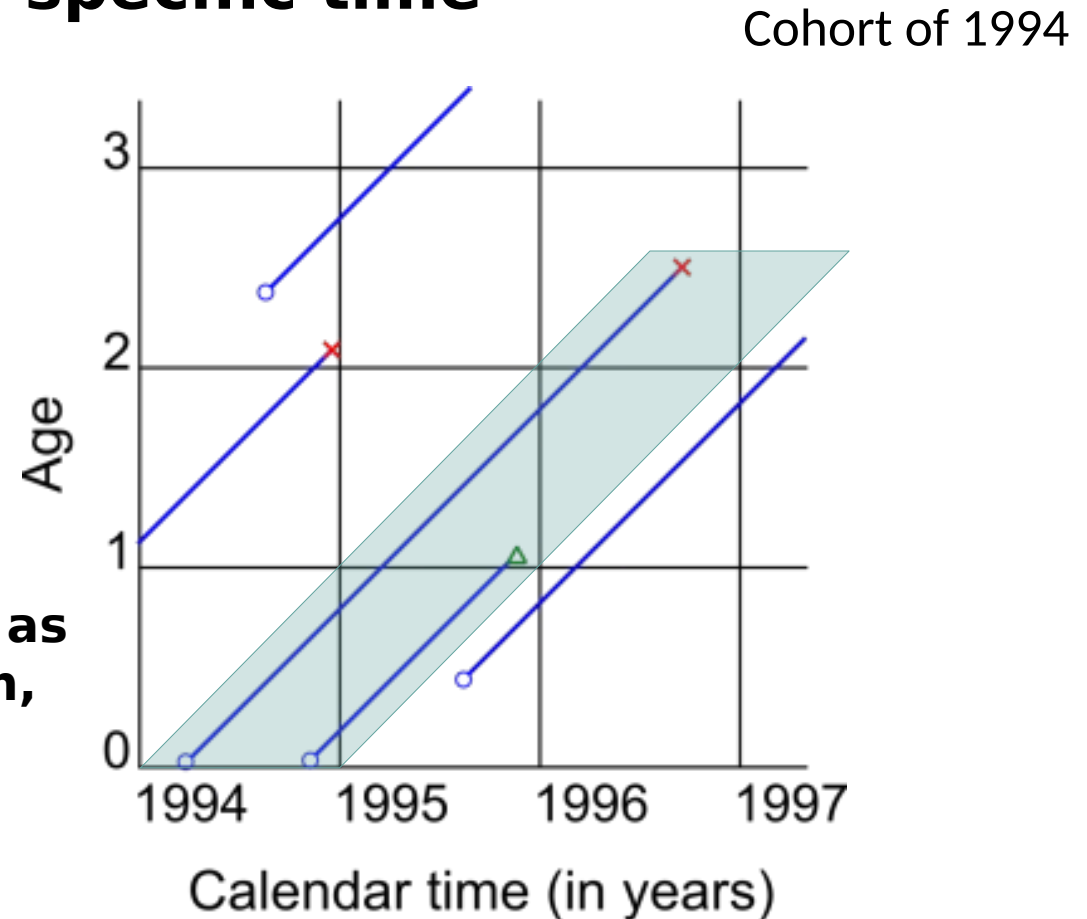
CONCEPT OF COHORT

Aggregate of all units (people) that experience a particular demographic event during a specific time interval

Eg: Indian Birth cohort of 1980

Cohort of FLAME class of 2025

Cohorts are commonly defined by **events such as births, marriages, puberty onset, migration, or first birth within a given period.**



UNDERSTANDING COHORT DATA

- Cohort analysis benefits from **offering detailed insights into the collective experiences of a group but faces feasibility challenges due to the long duration required for data collection.**
- To understand the fertility or mortality experience of a birth cohort, data collection can span decades, making it difficult to assess current cohorts comprehensively.
 - Historical cohort measures are more feasible because they rely on completed data, whereas partial cohort measures, which reflect a cohort's experience up to the present, remain inherently incomplete.
- **The distinctions between period measures and cohort measures can be more easily understood through visual aids such as Lexis diagrams,** which graphically represent these concepts.

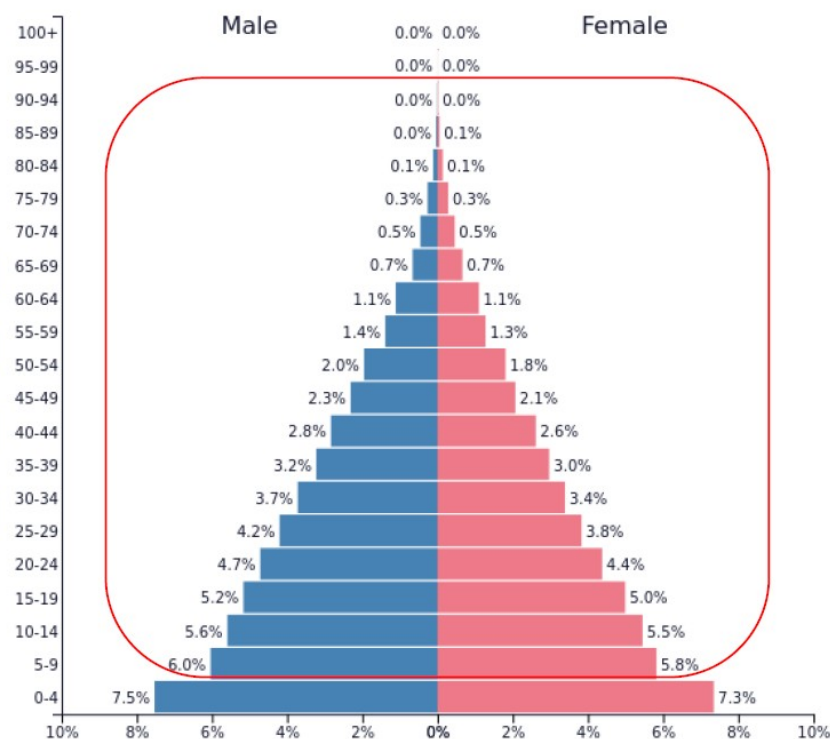
LITERACY

Table-25: Literacy Rates

Census year	Persons	Males	Females
1951	18.3	27.2	8.9
1961	28.3	40.4	15.4
1971	34.5	46.0	22.0
1981	43.6	56.4	29.8
1991	52.2	64.1	39.3
2001	64.8	75.3	53.7
2011	73.0	80.9	64.6

1951-1971: Age group 5 and above, 1981-2011: Age group 7 and above
Data Source: Office of the Registrar General & Census Commissioner, India

(In percentage)



PopulationPyramid.net

India - 1951
Population: 382,376,945

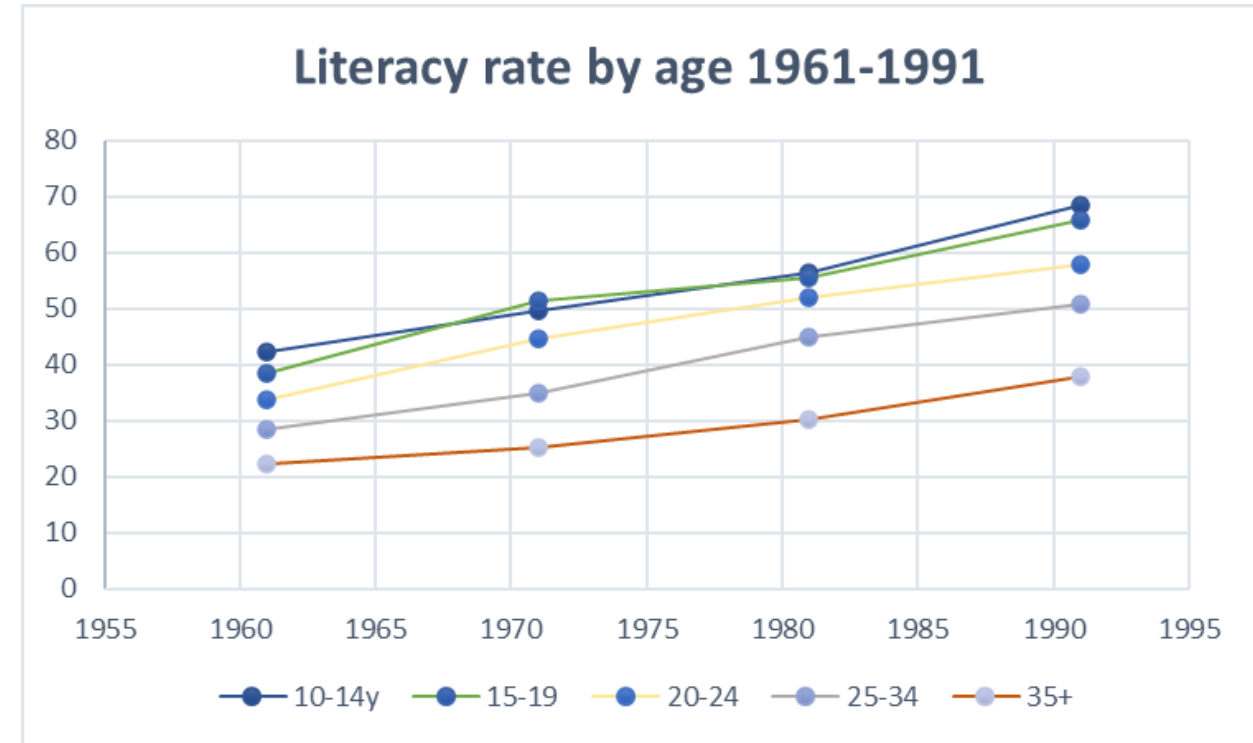
Source: MHRD Statistics



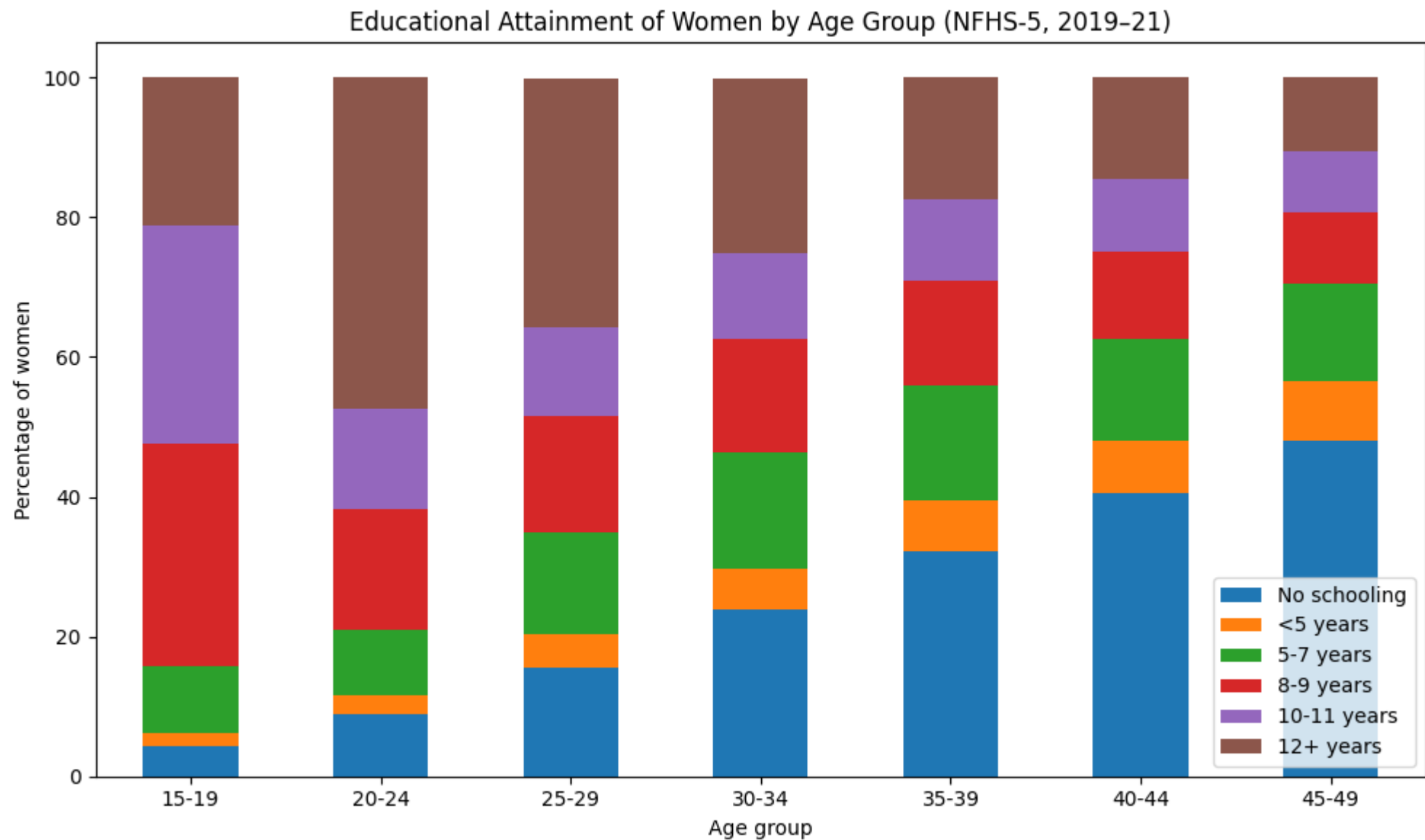
Table 5: Literacy rate in different age groups, 1961 - 1998/99

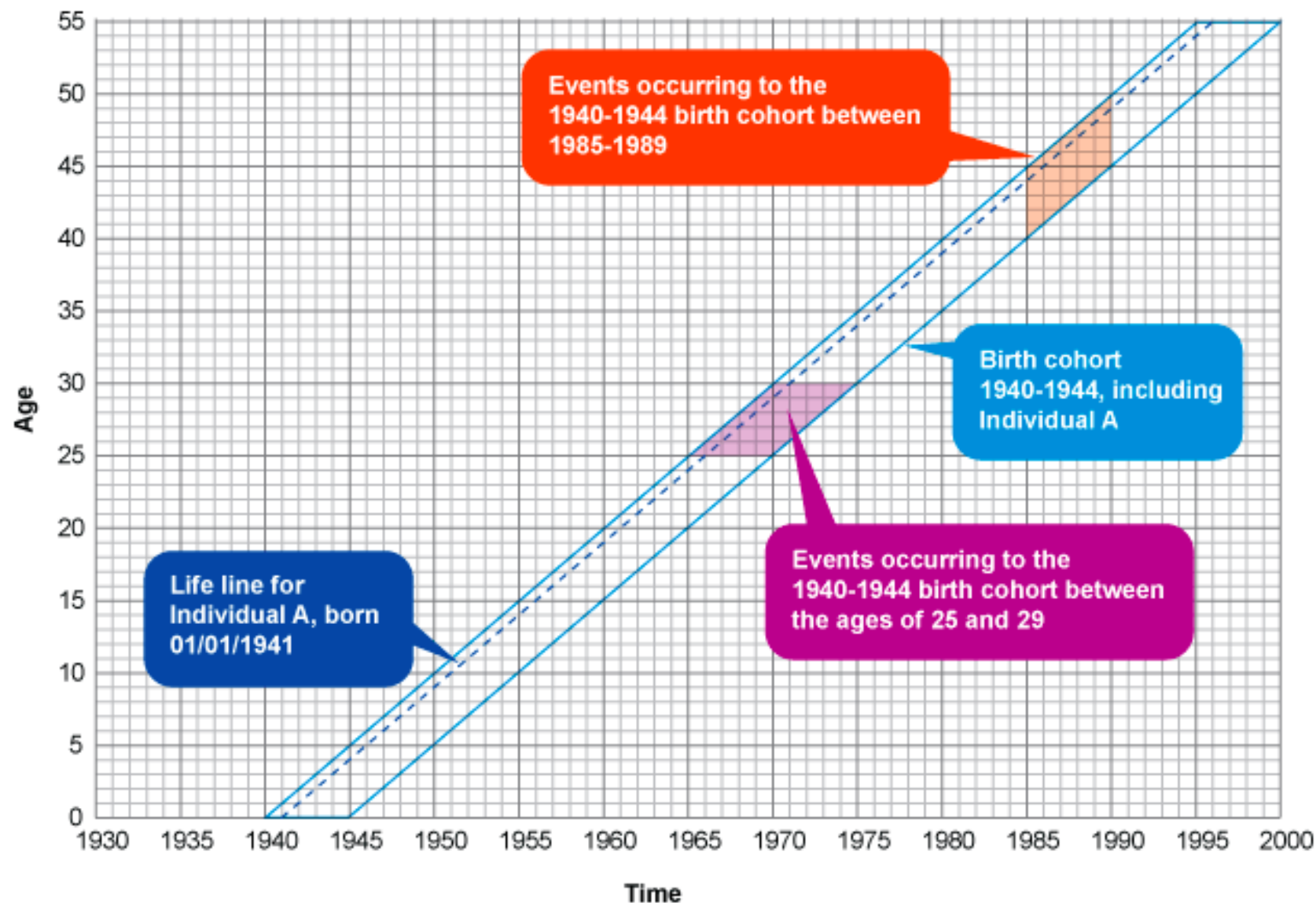
Age group	1961	1971	1981	1991	NFHS-2 1998-99
10-14	42.3	49.6	56.3	68.5	81.8
15-19	38.4	51.3	55.5	65.7	76.9
20-24	33.6	44.7	52.0	57.9	67.0 ^a
25-34	28.5	34.8	45.0	50.9	56.9 ^b
35+	22.2	25.2	30.3	38.0	45.2 ^c
15+	27.8	34.1	40.9	48.2	58.6

a) Age group: 20-29; b) Age group: 30-39; c) Age group: 40+



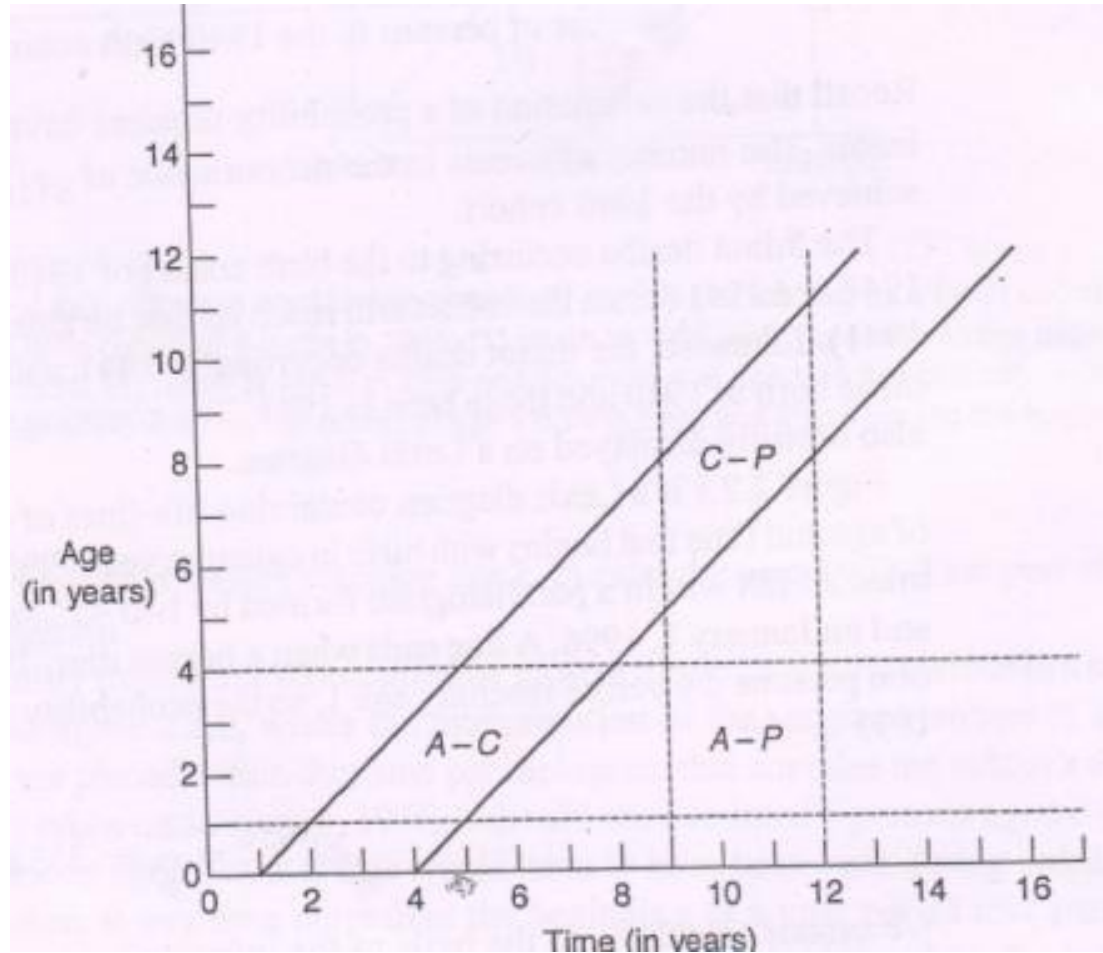
<https://www.educationforallinindia.com/page171.html>



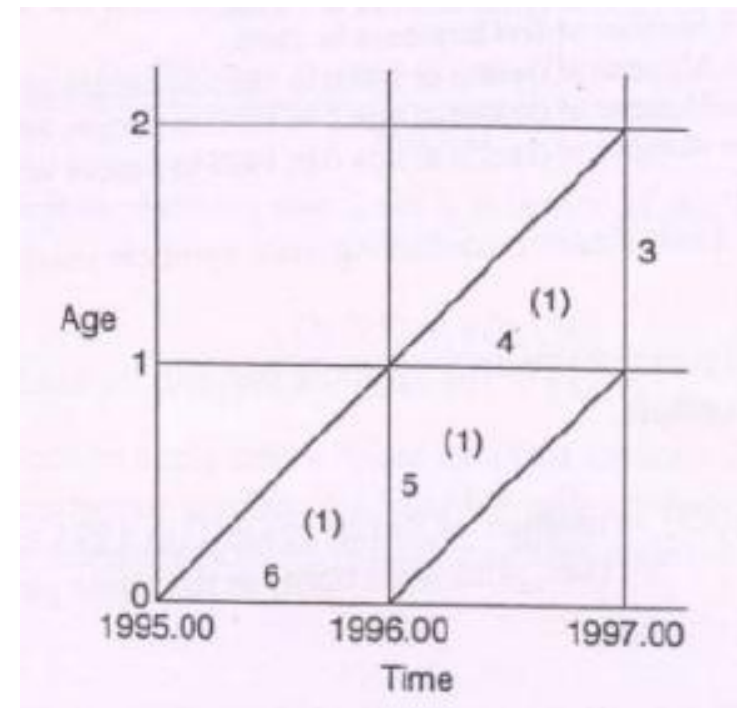
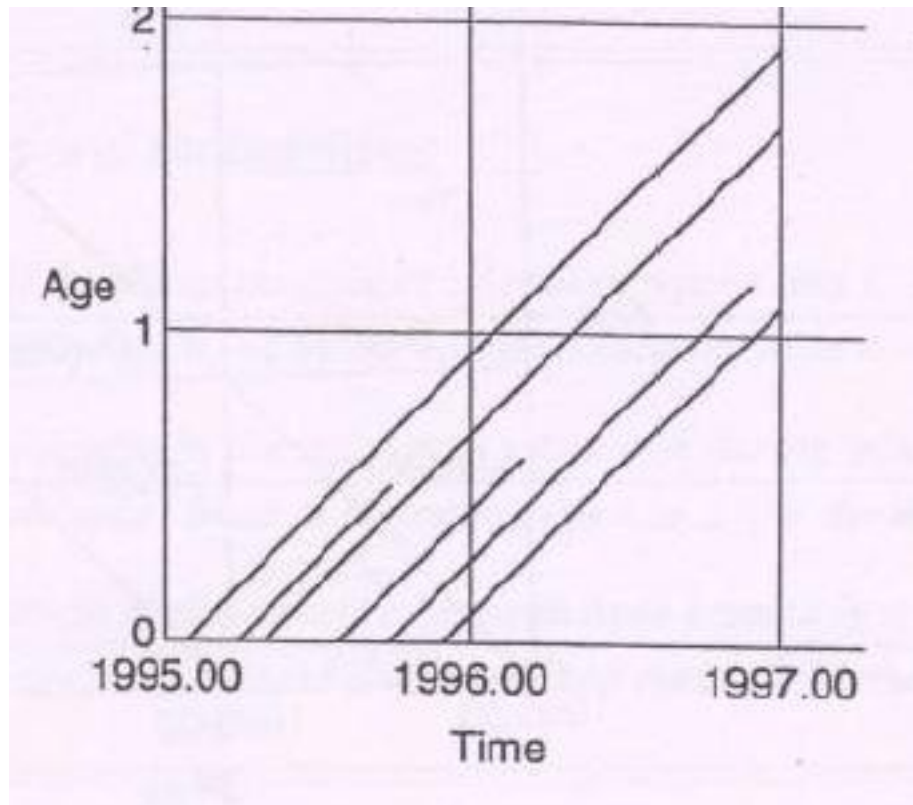


A cohort at a particular time (cohort-time measure) also can be seen as a "lozenge shape" which shows that the demographic events we are interested in are happening to a specific birth cohort in a specific time interval. For example, the 1940-44 birth cohort during the period 1985-1990 is shown as the orange "lozenge" shape. Notice how the shape is bounded by vertical lines which depict a time period.

LEXIS DIAGRAMS REVISITED



LEXIS DIAGRAMS- COHORT MORTALITY RATE



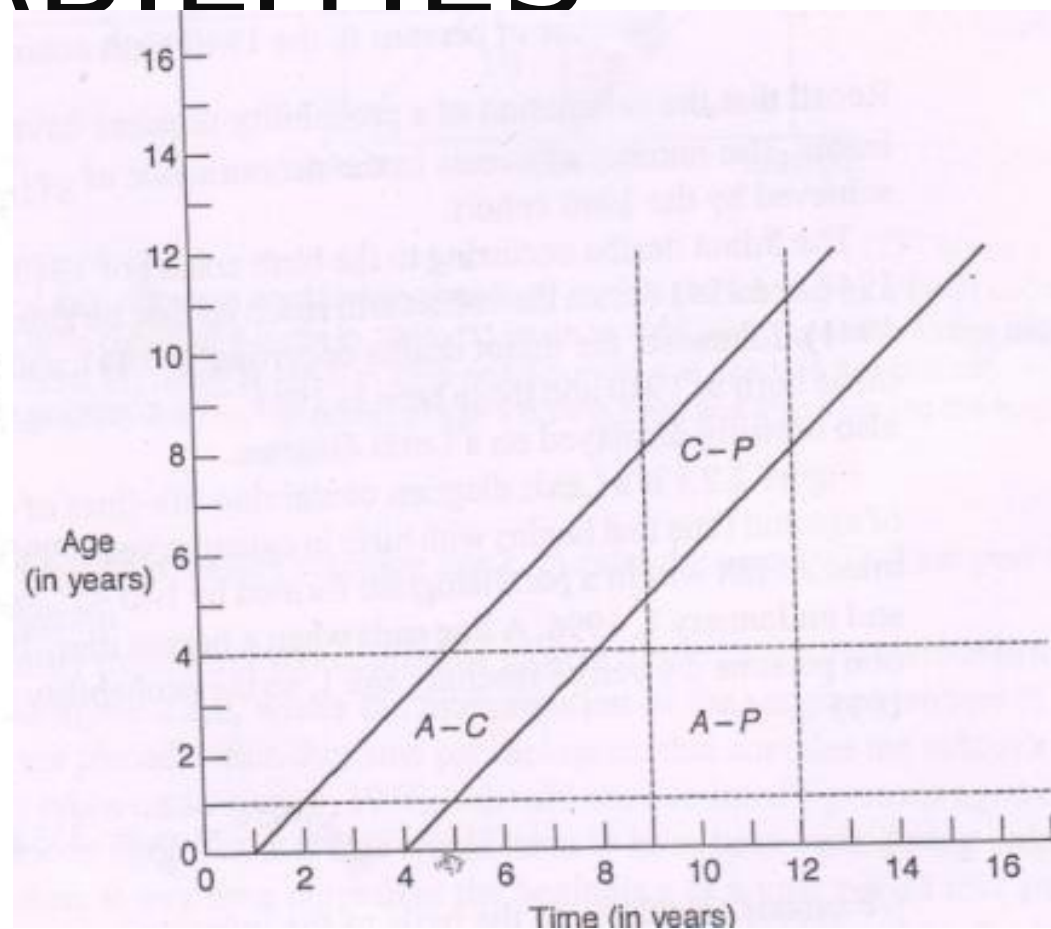
RATES AND PROBABILITIES

Box 2.3 Calculating Rates and Probabilities

$${}_nM_x^{\text{cohort } c} = \frac{\text{Number of deaths to the cohort } c \text{ between ages } x \text{ and } x + n}{\text{Number of person-years lived by the cohort } c \text{ between ages } x \text{ and } x + n}$$

$${}_nM_x^{\text{year } t} = \frac{\text{Number of deaths in the age range } x \text{ to } x + n \text{ during year } t}{\text{Number of person-years lived in the age range } x \text{ to } x + n \text{ during year } t}$$

$${}_nq_x^{\text{cohort } c} = \frac{\text{Number of deaths to the cohort } c \text{ between ages } x \text{ and } x + n}{\text{Number of persons in the cohort } c \text{ who reached their } x^{\text{th}} \text{ birthday}}$$



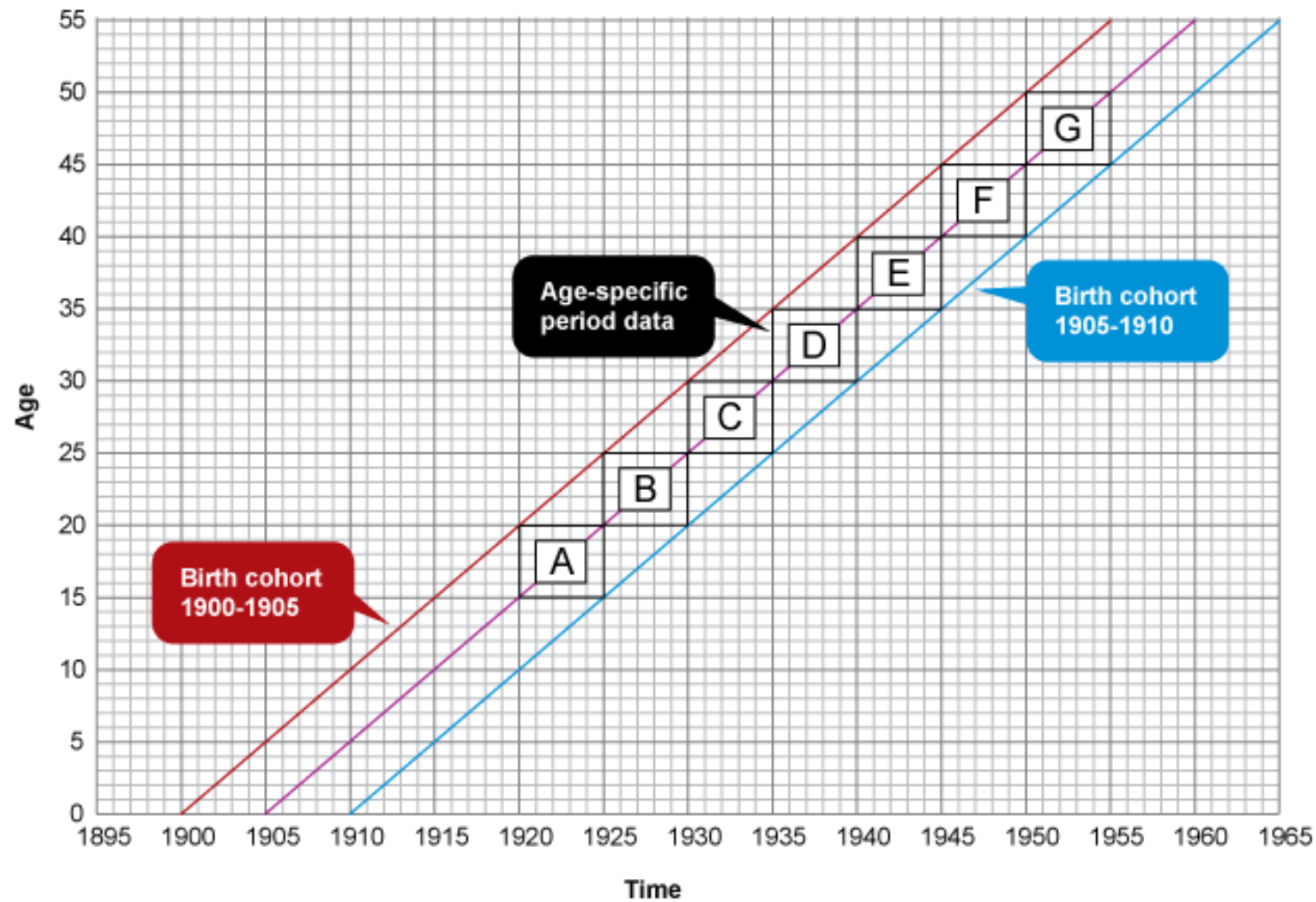
PERIOD AND COHORT DATA

Adapted from Pollard (1974)

Age Group	1920 - 24	1925 - 29	1930 - 34	1935 - 39	1940 - 44	1945 - 49	1950 - 54
15 - 19	0.026	0.029	0.026	0.025	0.024	0.030	0.039
20- 24	0.134	0.127	0.106	0.106	0.123	0.156	0.186
25 - 29	0.168	0.156	0.128	0.129	0.147	0.179	0.190
30- 34	0.138	0.123	0.100	0.097	0.109	0.126	0.124
35 - 39	0.098	0.086	0.065	0.058	0.062	0.074	0.066
40 - 44	0.041	0.035	0.027	0.021	0.020	0.023	0.021
45- 49	0.004	0.004	0.003	0.002	0.002	0.002	0.002

**FERTILITY
RATES**

http://papp.iussp.org/sessions/papp101_s02/PAPP101_s02_100_040.html#1



Nor does the recorded data correspond to a ten-year birth cohort. The sum of all the births in boxes A-G is not equal to the total births experienced by both birth cohorts. However, despite these downfalls, the translation of period into cohort data is still a useful approach with valid results.